

DASE7506 MP1 Individual Report

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Abstract

In this project, I trained a compact causal language model from random initialization using only the supplied WikiText-2 training text. In the original starter, `student.py` simply returned the unchanged classroom GPT. The submitted implementation replaces that placeholder with a separate `StudentGPT` while retaining the original baseline for comparison. The final predictor combines a dense decoder, auxiliary multi-token prediction during training, exponential moving average weights, matrix-only weight decay, a causal within-window neural cache, and a compact bigram distribution derived from the training split. Its complete test bits per byte (BPB) is 1.476789395, compared with 2.101260438 for the initial baseline, a 29.72% reduction. The final validation BPB is 1.459831448 and is reported separately from the test score.

On the measured laptop, median CPU FP32 scoring time is 4.398 times the paired baseline. Peak evaluation RAM is 1.888 GiB, and required uncompressed inference assets occupy 36.305 MiB. All three measurements satisfy the assignment limits. The final model received substantially more training than the baseline, so I also report an equal-target comparison, equal-budget weight-decay trials, and fixed-weight inference ablations.

1. Benchmark and Evaluation Protocol

The benchmark is protocol 7506-mp1-wt2-v2, using the supplied WikiText-2 raw text and train-fitted BPE tokenizer with vocabulary size 2,048. A direct comparison with the starter archive confirms that `model.py`, `common.py`, `evaluate.py`, `configs/baseline.json` and `tests/test_contract.py` remain byte identical. The course data and tokenizer also retain their release hashes. Development code restricts model training, calibration and statistical export to train and validation. No external training text, pretrained weights or evaluation network access is used.

Split	Scored targets	Raw UTF-8 bytes
Validation	376,599	1,148,007
Test	428,405	1,292,013

The independent causal windows of 256 targets are scored, including the final short window. Every target except the first token of each split is scored exactly once. Adjacent windows share one boundary token but carry no state. BPB equals summed next-token negative natural-log probability divided by $\ln(2)$ and by the split's complete raw UTF-8 byte length. Token perplexity is supplementary and is not comparable with published word-level perplexity.

Predictor	Validation BPB	Complete test BPB
Initial baseline	2.071083639	2.101260438
Final frozen predictor	1.459831448	1.476789395

2.Final Predictor

The original student factory returned the baseline GPT without modifying it. The final implementation preserves the same causal attention interface but changes the architecture, training objective and inference distribution.

Component	Original starter	Final submitted implementation
Scale	Width 128; 4 blocks; 4 heads	Width 224; 8 blocks; 7 heads; head dimension remains 32
Position encoding	Learned absolute embedding added to token embeddings	RoPE rotates queries and keys; deterministic tables are not stored in the checkpoint
Normalization	Pre-LayerNorm	Pre-RMSNorm
Feed-forward network	Four-times-width GELU MLP	SwiGLU with hidden width 600
Regularization and initialization	No model dropout; all linear outputs initialized at standard deviation 0.02	Embedding, attention and residual dropout 0.1; residual outputs use depth-scaled initialization
Training objective	Next-token cross-entropy only	Next-token loss plus two MTP heads for horizons 2 and 3, weighted by 0.2
Inference distribution	Direct log-softmax of neural logits	Temperature 1.11, 5% causal neural cache, then 2% training-derived bigram mixture
Preserved design	Causal scaled dot-product attention, tied token and output weights, bias-free output head	The same three properties are retained

The final neural model has 6,229,856 parameters during training and 5,312,352 after the auxiliary MTP heads are removed. RoPE removes the learned position table and injects position through query-key phase rotations. RMSNorm omits mean subtraction, while SwiGLU replaces the baseline GELU MLP with a gated feed-forward path. These are implementation differences, not independent causal findings: the available absolute-position trial also changed MTP and inference settings, so it is not a clean RoPE ablation.

At inference, the cache scores only earlier hidden states $i < t$ and copies an outcome token $\text{ids}[i+1]$ that is already present in the prefix. It resets for every window and example. The final bigram table contains 3,613,342 training pairs, uses $\alpha = 1$ smoothing, and contributes 2% probability mass. Its 16 MiB FP32 table contains no validation or test counts.

3. Training and Model Selection

Training setting	Original starter recipe	Final submitted trial
Budget	1,200 updates; 9,830,400 primary targets	24,000 updates; 196,608,000 primary targets
Batch and sampling	Seed 17; batch 32; random 257-token crops	Same seed, batch and random-crop policy
AdamW	Learning rate 0.001; beta2 0.999; weight decay 0.1	Learning rate 0.0006; beta2 0.95; matrix weight decay 0.6
Schedule	100-step warmup; cosine floor 10%	200-step warmup; cosine floor 5%
Parameter decay	All trainable parameters	Matrices only; biases and norm scales are exempt; tied weights are deduplicated
Checkpoint choice	Final training state	Complete validation every 500 steps; best raw or EMA state
Weight averaging	None	EMA 0.999 initialized at step 8,000; selected step 20,000
Development data access	load_data returned train, validation and test	Restricted loader returns train and validation only and rejects test requests
Training precision	Default CPU-oriented starter command	BF16 on NVIDIA RTX 3060 Laptop GPU; evaluation remains CPU FP32

Matrix-only weight decay includes the shared embedding matrix and auxiliary heads. Tied parameters are deduplicated. The reported target budget counts primary next-token targets; auxiliary targets add training work but are not added to this count. Equal primary target counts therefore do not imply equal FLOPs.

The selected checkpoint is the EMA state at step 20,000, not the final state at step 24,000. The complete trial still consumed 196,608,000 primary targets. The selected EMA gives 1.463201847 validation BPB with the original inference settings. The raw model reached its best logged value of 1.469339043 at step 23,500.

A refined validation-only grid evaluated 126 settings: the original configuration plus the Cartesian product of temperatures 1.10 to 1.14, cache weights 0.02 to 0.06, and theta values 12, 16, 20, 24 and 28. It selected temperature 1.11, cache weight 0.05 and theta 12, reducing validation BPB to 1.461640849. A separate 48-setting bigram grid selected alpha 1 and mixture weight 0.02. Export and the unchanged CPU FP32 evaluator confirmed 1.459831448. The final WD 0.6 candidate was selected on validation and froze before its reported test evaluation.

4. Training Curves

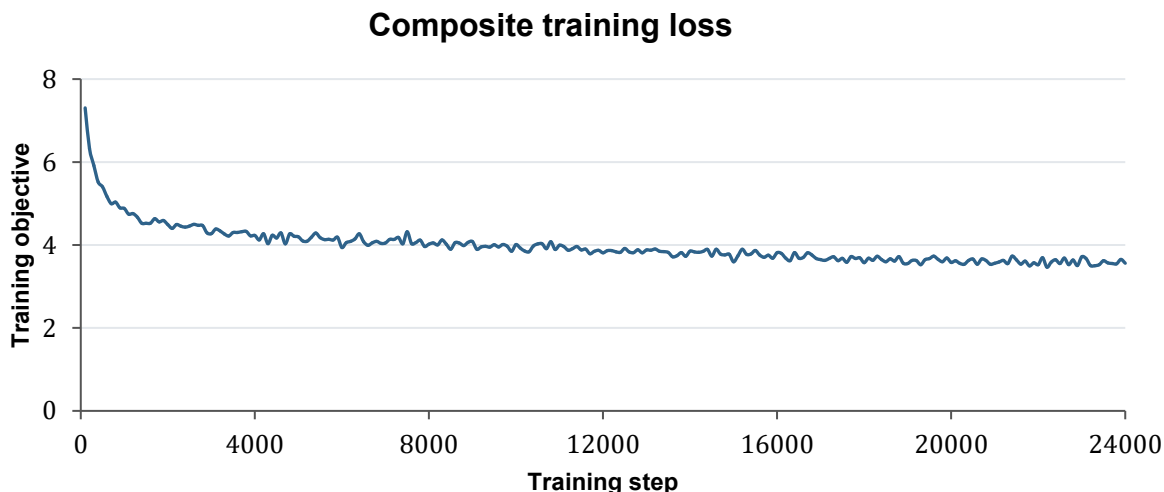


Figure 1. The final WD 0.6 trial records the current mini-batch composite loss every 100 steps, giving 240 real points. The objective includes primary next-token cross-entropy and the MTP auxiliary term. It is neither BPB nor a 100-step average. No smoothing is applied.

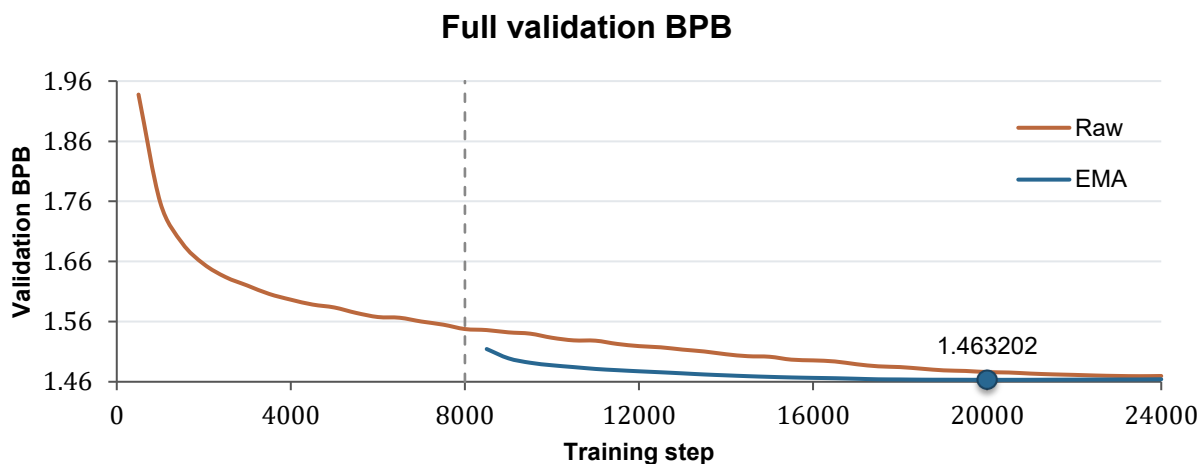


Figure 2. Full FP32 validation BPB from the same trial: 48 raw and 32 EMA observations. The dashed line marks EMA initialization at step 8,000; its first recorded evaluation is at step 8,500. The selected EMA at step 20,000 has BPB 1.463201847 with the original inference settings. The vertical axis does not start at zero.

Both charts use `runs/final-frozen-wd06-refined-bigram/training_metrics.json`. Lines connect recorded observations, not measured values at unlogged steps. The validation curve precedes refined calibration and bigram interpolation, so its minimum must not be labeled with the final predictor score of 1.459831448.

Late validation improvement slows and sometimes reverses. The selection rule therefore retains the best validation checkpoint rather than the last update. Both Word charts embed editable numerical workbooks containing the actual logged values.

5. Controlled Comparisons

Model	Primary targets	Parameters	Validation BPB
Baseline 128 by 4	9,830,400	1,088,256	2.071083639
Early student 128 by 4	9,830,400	1,059,136	1.840597866
Early student 192 by 6	9,830,400	3,061,824	1.732685562

All three runs used seed 17 and 1,200 updates. The early 128-by-4 student combined RoPE, RMSNorm and SwiGLU with a parameter count close to the baseline and reduced validation BPB by 0.230485773. Increasing width and depth to 192 by 6 improved validation further but also increased parameters. This comparison controls processed primary targets, not time, parameter count or FLOPs. It supports the combined design rather than isolating each architectural component.

Matrix WD	Selected EMA step	Comparable validation BPB
0.1	17,500	1.483033579
0.2	18,000	1.477100884
0.3	19,000	1.471285776
0.4	18,500	1.468591968
0.5	19,000	1.466051340
0.6	20,000	1.463201847

These six trials used the same 224-by-8 architecture, seed, sampling, learning-rate schedule, MTP settings, EMA policy and 24,000-update budget. Each processed 196,608,000 primary targets. The table uses the same original temperature and cache settings, so it is the cleanest available weight-decay comparison. Across these single-seed trials, increasing matrix decay from 0.1 to 0.6 reduced comparable validation BPB by 0.019831732. This supports 0.6 within the tested range; it does not show that still larger decay would help.

Postprocessing was selected separately on validation. The saved WD 0.5 pipeline reached 1.464390298 after bigram interpolation. The initial WD 0.6 pipeline reached 1.461203423, and the refined WD 0.6 calibration reduced the final pipeline to 1.459831448. Because the refined grid was applied only to the final candidate, this last comparison measures complete pipelines rather than weight decay alone.

For the earlier 192-by-8 MTP model, random-sampling runs at 12,000 and 16,000 steps achieved 1.511870852 and 1.500751094 validation BPB. EMA plus matrix-decay runs at 16,000 and 24,000 steps achieved 1.498867127 and 1.489091291. Changing total steps also changes the cosine schedule, so these are separate recipes rather than continuations of one trajectory.

6. Ablations and Unsuccessful Experiments

Fixed neural weights	Primary neural targets	Validation BPB
Refined neural model and window cache	196,608,000	1.461640849
Same model with 2% bigram	196,608,000	1.459831448

Bigram interpolation reduces BPB by 0.001809401. The exporter checks that every neural tensor remains bit-for-bit unchanged, so the paired difference isolates the training-derived statistical mixture rather than additional neural training. The extra work is counting 3,613,342 training pairs and searching 48 validation candidates. The 16 MiB table accounts for most non-neural inference storage, while the BPB gain is small relative to the overall improvement over the baseline.

The coarse calibration grid provides a fixed-weight cache ablation for the final WD 0.6 neural checkpoint. With temperature 1.12, the best no-cache candidate gives 1.485866321 validation BPB. Cache weight 0.04 and theta 20 give 1.463033365, an improvement of 0.022832956 using the same neural weights. The refined grid later selected temperature 1.11, cache weight 0.05 and theta 12. The cache gain comes with additional similarity computation and is the main reason CPU evaluation approaches the five-times limit.

Trial	CPU validation BPB	Interpretation
192 by 8 MoE4 at 16,000 steps	1.565672484	Tested token-routing design was not retained
Absolute positions at 8,000 steps	1.580118982	Exploratory trial also changed MTP and cache settings; not a clean RoPE ablation
Epoch sampling with MTP at 8,000 steps	1.544776716	Random sampling gave 1.535001231
MTP weight 0.1 with EMA at 16,000 steps	1.501337150	Weight 0.2 gave 1.499784120
MTP weight 0.4 with EMA at 16,000 steps	1.505492763	Stronger auxiliary loss did not help
Nine-block WD 0.5 model	1.466591001	Deeper model was worse than the eight-block control
WD 0.5 and WD 0.6 weight soup	1.463201823	Worse than the selected WD 0.6 endpoint

These trials document the search rather than prove that an architecture is generally inferior. Several comparisons change more than one factor or use different budgets. There is no fully matched MTP-disabled experiment, and the early comparison does not isolate normalization, position encoding and feed-forward design. The principal controlled mechanism ablations are the fixed-weight bigram and cache comparisons.

7.Resource Limits and Recorded Costs

Constraint	Measurement	Limit	Result
CPU scoring time	Median 24.743854 s; baseline 5.626390 s	At most 5 times	4.397821 times
Peak evaluation RAM	2,027,118,592 bytes; 1.887901 GiB	4 GiB	Pass
Uncompressed inference assets	38,069,018 bytes; 36.305445 MiB	64 MiB	Pass

The acceptance measurement used three paired complete-test runs on the same Intel Core i7-12700H laptop, Windows 11, Python 3.12.14, CPU FP32 and four threads. Each scorer started in a fresh process. Timing uses unchanged evaluate.py scoring seconds. Peak RAM is a conservative sum of per-process Windows PeakWorkingSetSize maxima for the scorer and descendants, queried every 50 ms. It includes loading and the Python launcher but excludes the parent measurement helper.

Required inference assets are checkpoint.pt, student.py and student_bigram.py, including the persistent bigram buffer. The size is uncompressed. Historical runs are excluded. The fixed evaluator, course data and software dependencies are necessary for reproduction but are not learned inference assets.

Final training or search stage	Recorded seconds	Scope
WD 0.5 neural training	1,652.554	Training intervals excluding intermediate validation
WD 0.5 complete process	1,804.198	Preparation, training and validation
WD 0.6 neural training	1,598.499	Training intervals excluding intermediate validation
WD 0.6 complete process	1,744.669	Preparation, training and validation
WD 0.6 intermediate validation	137.422	Raw and EMA validation
Refined cache and temperature grid	5.648	126 candidates; shared CUDA FP32 features
Bigram grid	25.138	48 validation candidates; CPU FP32
Bigram export	5.593	No neural training

The table summarizes the recorded computational cost of the final training and model-selection stages. Neural optimization dominated the total cost: the complete WD 0.5 and WD 0.6 runs took 1,804.198 and 1,744.669 seconds, respectively, including preparation, training, and intermediate validation. For the final WD 0.6 run, 1,598.499 seconds were spent on parameter updates and 137.422 seconds on raw and EMA validation. In comparison, validation-only post-processing was inexpensive: the refined cache and temperature search over 126 candidates took 5.648 seconds, the 48-candidate bigram search took 25.138 seconds, and exporting the selected bigram table took 5.593 seconds without additional neural training. The neural-training, intermediate-validation, and complete-process measurements overlap and should therefore not be added together.

8. Reproduction and Provenance

Run these PowerShell commands from code/. Direct evaluation needs neither a GPU nor retraining. The tested environment used Windows 11, Python 3.12, PyTorch 2.7.1 and four CPU threads.

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install torch==2.7.1 `
  --index-url https://download.pytorch.org/whl/cpu
python -m pip install -r requirements-eval.txt
python -m unittest discover -s tests -v
python evaluate.py `
  --checkpoint runs/final-frozen-wd06-refined-bigram/checkpoint.pt `
  --device cpu --precision fp32 --threads 4 --split test `
  --output runs/final-frozen-wd06-refined-bigram/reproduced_test.json
```

The expected complete-test result is BPB 1.476789395067789, with 428,405 targets and 1,292,013 UTF-8 bytes. Small FP32 differences may occur across platforms. Retraining is unnecessary for evaluation and does not guarantee a byte-identical checkpoint.

```
python train.py --implementation student `
  --config configs/student_224x8_mtp2.json `
  --device cuda --precision bf16 --threads 4 --seed 17 `
  --steps 24000 --batch-size 32 --sampling random `
  --learning-rate 0.0006 --warmup-steps 200 `
  --min-lr-ratio 0.05 --weight-decay 0.6 `
  --weight-decay-policy matrix --adam-beta2 0.95 `
  --ema-decay 0.999 --ema-start-step 8000 `
  --eval-every 500 --keep-best --run-dir runs/repro-wd06
```

Validation-only postprocessing uses temperatures 1.10 to 1.14, cache weights 0.02 to 0.06, theta values 12, 16, 20, 24 and 28, followed by tune_bigram.py and export_bigram.py with alpha 1 and weight 0.02. A fresh training run may select different validation settings.

File	SHA256
checkpoint.pt	c8c30df36e1e0b29beb11c36155d3f36020d2f8ad8462ee55d89e25a77db574f
student_bigram.py	689c6a9c2b1934687c2eaec4f06593c3e604f098f7c16e8afbe247903e334b7c
student.py	71a43d118b593b0e253eb3d49d767f965e353632604e625f991843d3823f9022
evaluate.py	128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d
tokenizer.json	020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e

Evidence is archived under runs/final-frozen-wd06-refined-bigram/, including training, calibration, bigram search and export records, validation and test scorer outputs, the frozen manifest, and acceptance-final/acceptance.json. The starter's original contract test is unchanged. Additional tests cover the development-data lock, calibration, bigram normalization, dropout compatibility, EMA, matrix-only decay, weight averaging and width or depth variants; all 26 current tests pass.

9. Discussion and Disclosure

The final complete-test BPB is 1.476789395, and all three measured resource limits pass. Equal-target early comparisons support the combined student architecture. Six equal-budget trials support matrix weight decay 0.6 within the tested range. Fixed-weight cache and bigram ablations show measurable validation gains without changing the neural weights. The overall baseline-to-final improvement combines architecture, training budget, optimization, checkpoint averaging and inference mechanisms, so it cannot be assigned to one component.

All main configuration trials use seed 17, so the report provides no multi-seed confidence intervals. Repeated validation searches can make the selected validation result optimistic. The validation-to-test gap is 0.016957947 BPB. CPU timing has limited headroom below the five-times ceiling and should be checked during independent reproduction.

This work reuses the course baseline, data processing, tokenizer and evaluator. Standard mechanisms are attributed below and are not claimed as original algorithms. I executed the full training runs locally and remains responsible for the implementation and claims.

References

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